

# ALPK3-Related Hypertrophic Cardiomyopathy: Comprehensive Disease Report

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## Summary

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**ALPK3-related hypertrophic cardiomyopathy (HCM)** is a Mendelian sarcomeric/M-band cardiomyopathy caused by loss-of-function (LoF) variants in *ALPK3* (alpha-protein kinase 3), located at chromosome **15q25.3** (HGNC:17870; NCBI Gene 57538; OMIM gene 617608; disease MIM 618052, **CMH27**). *ALPK3* encodes a **catalytically-dead atypical  $\alpha$ -kinase** that functions as a **structural M-band scaffold** of the cardiac sarcomere, anchoring myomesins (MYOM1/2), MuRF (muscle ring-finger) E3 ubiquitin ligases, and the autophagy adaptor SQSTM1/p62 to maintain thick-filament proteostasis. It is a distinctive disease because it exhibits a **dose-dependent, zygoty-driven severity gradient**: biallelic (recessive) LoF causes a severe, often lethal prenatal/pediatric cardiomyopathy with syndromic extracardiac features, whereas heterozygous (dominant) protein-truncating variants (*ALPK3*tv) cause an incompletely penetrant adult-onset HCM that accounts for roughly **1–4% of adult HCM**.

The core mechanism is now well supported: *ALPK3* loss displaces myomesins from the M-band, driving **thick-filament protein aggregation, sarcomere and intercalated-disc disarray, abnormal calcium handling, and hypercontractility**. In model systems, these defects were partially corrected by the myosin inhibitor **mavacamten**, and durable phenotypic rescue was achieved in global knockout mice using **AAV-delivered ALPK3**, establishing two genotype-directed therapeutic proof-of-concept fronts that remain preclinical. Clinically, the adult heterozygous phenotype has a characteristic morphology — **apical/septal hypertrophy, apical aneurysm, right-ventricular involvement, and myocardial fibrosis** — and *ALPK3* has emerged as one of the leading (frequently the second most common) genotypes identified in apical HCM cohorts.

Management currently follows **standard HCM guidelines** (beta-blockers, disopyramide, septal reduction therapy, ICD for sudden-cardiac-death [SCD] prevention, and transplantation for end-stage disease), with **cardiac MRI late gadolinium enhancement (LGE)** serving as a key risk-stratification marker particularly relevant to the fibrosis- and aneurysm-prone *ALPK3*

phenotype. ALPK3 now carries **established gene–disease validity for HCM with recognized dual (recessive and dominant) inheritance** per the ClinGen Hereditary Cardiovascular Disease Gene Curation Expert Panel.

## 1. Disease Information

**Overview.** ALPK3-related cardiomyopathy is an inherited cardiac muscle disease presenting predominantly as hypertrophic cardiomyopathy, but with a broad phenotypic spectrum that also includes dilated cardiomyopathy (DCM), a neonatal DCM-to-hypertrophy transition, and restrictive/mixed physiology. The disease exists as a **spectrum defined by zygosity**: severe recessive pediatric disease at one pole and penetrance-limited adult dominant HCM at the other.

### Key identifiers.

| Resource          | Identifier                                                      |
|-------------------|-----------------------------------------------------------------|
| Gene symbol       | <i>ALPK3</i> (α-protein kinase 3; formerly <i>MIDORI</i> )      |
| HGNC              | HGNC:17870                                                      |
| NCBI Gene         | 57538                                                           |
| Cytogenetic locus | 15q25.3                                                         |
| OMIM (gene)       | 617608                                                          |
| OMIM (disease)    | 618052 — Cardiomyopathy, familial hypertrophic, 27 (CMH27)      |
| MONDO             | Cardiomyopathy, familial hypertrophic, 27 (maps to OMIM 618052) |
| ICD-10            | I42.1 (obstructive HCM) / I42.2 (other HCM)                     |
| ICD-11            | BC43.0 (hypertrophic cardiomyopathy)                            |
| MeSH              | D002312 (Cardiomyopathy, Hypertrophic)                          |

**Synonyms / alternative names:** ALPK3 cardiomyopathy; ALPK3-related HCM; CMH27; alpha-kinase 3 cardiomyopathy; biallelic ALPK3 pediatric cardiomyopathy (recessive form); ALPK3tv HCM (heterozygous truncating-variant form).

**Information source.** Evidence is derived from **aggregated disease-level and patient-level literature** — systematic variant curations, multicentric cohorts (French, Swedish, Chinese), individual case reports, and functional model-organism/iPSC studies — rather than from a single EHR dataset.

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## 2. Etiology

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**Primary causal factor: genetic.** ALPK3-related cardiomyopathy is a **monogenic disorder** caused by loss-of-function variants in *ALPK3*. There is no established environmental or infectious cause. The disease displays a **dose-dependent relationship between ALPK3 gene dosage and phenotype severity** (Finding F001; Finding F007).

**Genetic risk factors.** - **Biallelic LoF variants** (homozygous or compound heterozygous nonsense, frameshift, or splice variants) → severe, often lethal, prenatal/early-onset cardiomyopathy with extracardiac involvement (recessive). - **Heterozygous protein-truncating variants** (nonsense/frameshift/splice) → adult-onset dominant HCM explaining ≈1–4% of adult HCM. - As stated in the integrative review ([PMID: 41221624](#)): *"Biallelic loss-of-function variants lead to severe, often lethal cardiomyopathy with prenatal or early onset presentation and extracardiac involvement. Heterozygous protein-truncating variants, defined as nonsense or frameshift (resulting from insertion/deletion events or splicing mutations), explain ≈1% to 4% of adult hypertrophic cardiomyopathy."*

**Environmental / demographic risk factors.** No specific toxic, occupational, or infectious triggers are established. **Male predominance** is reported among heterozygous ALPK3tv HCM patients, and **age** is a strong modifier given the late onset and age-dependent penetrance ([PMID: 41645375](#)). **Consanguinity** is an important contextual factor for the recessive form — the founding families were consanguineous ([PMID: 26846950](#)).

**Protective factors.** No specific genetic or environmental protective alleles are established for ALPK3 cardiomyopathy. The **incomplete penetrance** of heterozygous variants (~20% in the founding family; even lower in later cohorts) implies the existence of unidentified genetic/environmental modifiers that buffer disease expression.

**Gene–environment interactions.** Not specifically characterized. The variable expressivity within families carrying identical variants (e.g., a mother with severe obstructive HCM and an asymptomatic brother carrying the same variant; [PMID: 41645375](#)) points to modifier effects, but specific gene–environment interactions have not been mapped.

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### 3. Phenotypes

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ALPK3 cardiomyopathy has distinct **cardiac** and **extracardiac** phenotype clusters, with the extracardiac cluster largely confined to the biallelic (recessive) pediatric form.

## Cardiac phenotypes

| Phenotype                                   | Type                   | HPO term (suggested)        | Onset                                        | Frequency / notes                                                                                  |
|---------------------------------------------|------------------------|-----------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------|
| Hypertrophic cardiomyopathy                 | Clinical sign          | HP:0001639                  | Neonatal (biallelic) to adult (heterozygous) | Predominant phenotype; 26/31 adults in French cohort                                               |
| Dilated cardiomyopathy / neonatal DCM       | Clinical sign          | HP:0001644                  | Neonatal                                     | 8/18 (44.4%; 95% CI 21.5–69.2%) biallelic live-borns had neonatal DCM transitioning to hypertrophy |
| Apical hypertrophy                          | Physical manifestation | HP:0001639 (apical variant) | Adult                                        | Frequent in heterozygous ALPK3tv; ALPK3 a leading ApHCM genotype                                   |
| Apical aneurysm                             | Physical manifestation | HP:0025169 (LV aneurysm)    | Adult                                        | ~17.2% in ApHCM cohort                                                                             |
| Right ventricular involvement               | Physical manifestation | HP:0001707                  | Adult                                        | Reported in heterozygous form                                                                      |
| Myocardial fibrosis (LGE)                   | Laboratory/imaging     | HP:0001637                  | Adult                                        | Prominent; prognostically important                                                                |
| Left ventricular systolic dysfunction       | Clinical sign          | HP:0005162                  | Pediatric                                    | E.g., patient 2, <a href="#">P 40447126</a>                                                        |
| Restrictive physiology / hypertrabeculation | Clinical sign          | HP:0001723                  | Adult                                        | Overlap phenotype ( <a href="#">P 36660067</a> )                                                   |
| Atrial fibrillation                         | Clinical sign          | HP:0005110                  | Adult                                        | 41.4% in ApHCM cohort                                                                              |

| Phenotype                                 | Type             | HPO term (suggested) | Onset | Frequency / notes                            |
|-------------------------------------------|------------------|----------------------|-------|----------------------------------------------|
| Non-sustained ventricular tachycardia     | Clinical sign    | HP:0004758           | Adult | 29.3% in ApHCM cohort                        |
| Sudden cardiac death / HCM-related events | Clinical outcome | HP:0001645           | Any   | 36.2% had HCM-related events in ApHCM cohort |

### Extracardiac phenotypes (biallelic/recessive form — syndromic)

| Phenotype                      | HPO term (suggested)    | Notes                                                    |
|--------------------------------|-------------------------|----------------------------------------------------------|
| Facial dysmorphism             | HP:0001999              | Recurrent triad component                                |
| Cleft palate                   | HP:0000175              | Reported in Tunisian case ( <a href="#">P 30046096</a> ) |
| Pectus excavatum / carinatum   | HP:0000767 / HP:0000768 | Thoracic deformity                                       |
| Scoliosis                      | HP:0002650              | <a href="#">P 40447126</a>                               |
| Joint contractures             | HP:0002803              | <a href="#">P 40447126</a>                               |
| Short stature                  | HP:0004322              | Multiple biallelic cases                                 |
| Webbed neck                    | HP:0000465              | <a href="#">P 40447126</a>                               |
| Hand/foot skeletal deformities | HP:0001155 / HP:0001760 | <a href="#">P 30046096</a>                               |

**Onset, severity, progression:** Biallelic disease is **neonatal/prenatal-onset, severe, and often lethal** (progressive; frequent early death). Heterozygous disease is **adult-onset (delayed ~10 years vs. sarcomeric HCM), variable severity, slowly progressive** with age-dependent

penetrance. **Quality-of-life impact** ranges from catastrophic (perinatal heart failure, death, transplantation in the biallelic form) to variable in adults (exertional symptoms, arrhythmia burden, ICD, and SCD risk).

## 4. Genetic / Molecular Information

**Causal gene.** *ALPK3* (HGNC:17870; NCBI Gene 57538; OMIM 617608), 15q25.3. Encodes alpha-protein kinase 3, an atypical  $\alpha$ -kinase that is **catalytically dead** and functions structurally.

**Pathogenic variant spectrum.** The disease is driven overwhelmingly by **truncating / loss-of-function** variants: - **Nonsense** (e.g., p.Arg1173, p.Ser653) - **Frameshift** (e.g., c.1531\_1532delAA p.Lys511Argfs12; c.109del p.R37Gfs72; c.2757dup p.T920Hfs14; c.3272del p.G1091Vfs43; c.1550dupC p.Pro518ThrfsTer53) - **Splice-site** variants - **Contiguous gene deletion** of 15q25.2q25.3 has been reported in a biallelic context

([P 37671554](#))

- Some **missense** variants (e.g., p.Arg1164Gln) appear in overlap/adult phenotypes

([P 36660067](#))

Documented patient-level variants exceed 150 in systematic curation (156 variants; Finding F001).

**Variant classification (ACMG/AMP).** Truncating variants are generally classified **Pathogenic / Likely Pathogenic**; classification for heterozygotes was historically complicated because *ALPK3* was initially annotated as autosomal recessive — re-analysis after dominant inheritance was established upgraded several variants

([P 41645375](#))

Bayesian/segregation evidence has supported Likely Pathogenic calls for recurrent truncating variants.

**Allele frequency.** Heterozygous *ALPK3* truncating variants are rare but present in population databases (gnomAD); their appreciable heterozygous frequency combined with incomplete penetrance underlies the ~1–4% contribution to adult HCM.

**Somatic vs. germline.** All disease-associated variants are **germline**. No somatic mechanism is implicated.



**Functional consequence. Loss of function** (haploinsufficiency in heterozygotes; complete loss in biallelic patients) — loss of the M-band scaffolding function of ALPK3.

**Modifier genes / epigenetics.** No specific Mendelian modifier genes are established. Notably, **microRNA-384-5p** regulates the ALPK3 pathway in hypertrophy models (Finding F009): *"miR-384-5p was notably decreased in cardiac hypertrophic tissues and cells, and overexpression of miR-384-5p could ameliorate pressure overload"* (PMID: 35510648). This provides an epigenetic/post-transcriptional regulatory axis. Chromosomal abnormalities are limited to the reported 15q25 contiguous deletion.

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## 5. Environmental Information

ALPK3 cardiomyopathy is a **monogenic disease with no established environmental, lifestyle, or infectious etiology**. No toxins, radiation, pollutants, dietary factors, or pathogens are implicated as causes or triggers. As with HCM generally, **strenuous competitive exercise** may modulate SCD risk and symptom burden, and standard HCM lifestyle counseling applies, but these are not ALPK3-specific causal factors. Age and male sex act as demographic modifiers of penetrance in the heterozygous form (see Section 2).

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## 6. Mechanism / Pathophysiology

### Ordered causal chain

1. **Loss-of-function variant in ALPK3** (biallelic → complete loss; heterozygous → haploinsufficiency) **leads to** reduced/absent functional ALPK3 protein at the sarcomeric M-band.
2. Loss of ALPK3 **results in** failure of its scaffolding function — it can no longer anchor **myomesins (MYOM1/2), MuRF E3 ubiquitin ligases, and SQSTM1/p62** at the M-band.
3. Displacement of myomesins and loss of the proteostasis hub **leads to** impaired thick-filament protein turnover and **thick-filament protein aggregation**.
4. Aggregation and scaffold failure **result in disordered sarcomeres and intercalated discs** (ultrastructural disarray).

5. Sarcomere disarray **leads to abnormal calcium handling** (elevated diastolic calcium) and altered myosin mechanochemistry — a **reduced super-relaxed (SRX) myosin fraction**, prolonged relaxation, and **hypercontractility** (demonstrated in the K201X knock-in mouse).
6. Reduced PKA phosphorylation of cardiac troponin I and impaired relaxation **result in diastolic dysfunction and contractile impairment**.
7. Chronic contractile/proteostatic stress **leads to myocyte hypertrophy, myocardial fibrosis, and (in neonates) a dilated-to-hypertrophic transition** → clinical HCM/DCM, **apical aneurysm, arrhythmia, heart failure, and SCD risk**.

*Branch:* In the **neonatal biallelic** context, the initial manifestation is often **dilated cardiomyopathy that subsequently transitions to hypertrophy** — a developmentally distinct branch, consistent with ALPK3's essential role in early cardiomyocyte differentiation (PMID: 26846950; PMID: 32480058).

*Inferred vs. demonstrated:* Steps 2–5 are **directly demonstrated** in iPSC-CM/ESC-CM models and mouse knock-ins

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P 27106955

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The downstream fibrosis/arrhythmia/SCD steps (step 7) are **inferred** from clinical cohorts and general HCM pathophysiology.

## Supporting detail

**Molecular pathways & protein dysfunction.** ALPK3 is a **catalytically-dead M-band scaffold** (Finding F002): "*ALPK3 lacks catalytic activity and maintains sarcomeric proteostasis by scaffolding MYOMs (myomesins), MuRF (muscle ring-finger protein) E3 ligases, and SQSTM1 (sequestosome-1)/p62. Loss of this scaffolding function displaces MYOMs, drives thick-filament protein aggregation, and precipitates severe contractile dysfunction*" (PMID: 41221624). The relevant pathway is **sarcomere/myofilament assembly and the ubiquitin–proteasome/autophagy quality-control system** (MuRF ligases, p62). The founding study additionally described ALPK3 as a **nuclear kinase essential for early cardiomyocyte differentiation** (PMID: 26846950), and the  $\alpha$ -kinase-3 mouse work supports a **scaffold model recruiting machinery for thick-filament protein turnover** (PMID: 40135575).

**Cellular processes.** Disrupted **sarcomeric proteostasis, autophagy/protein turnover, and calcium homeostasis** in cardiomyocytes. Ultrastructural evidence: *"Ultra-structural analysis of cardiomyocytes derived from patient-specific and human ESC-derived stem cell lines lacking ALPK3 revealed disordered sarcomeres and intercalated discs"* (PMID: 27106955).

**Metabolic/biochemical abnormalities.** Reduced SRX myosin fraction (an energy-conserving myosin state), elevated diastolic calcium, and reduced PKA-mediated troponin I phosphorylation — collectively producing **hypercontractility and impaired relaxation** (PMID: 40128237).

**Tissue damage.** Chronic myofibrillar stress → **myocardial fibrosis** (prominent LGE), apical aneurysm formation, and arrhythmogenic substrate.

**Suggested ontology terms.** - GO biological process: sarcomere organization (GO:0045214); myofibril assembly (GO:0030239); cardiac muscle hypertrophy (GO:0003300); regulation of protein catabolic process (GO:0042176); regulation of the force of heart contraction (GO:0002026); cardiac muscle cell differentiation (GO:0055007). - GO cellular component: M band (GO:0031430); sarcomere (GO:0030017); myofibril (GO:0030016); nucleus (GO:0005634). - CL cell types: cardiac muscle cell / cardiomyocyte (CL:0000746); regular cardiac myocyte (CL:0002098).

## 7. Anatomical Structures Affected

**Organ level.** Primary organ: the **heart** (UBERON:0000948), specifically the **cardiac ventricles** — left ventricle (UBERON:0002084) with frequent **apex** and **interventricular septum** involvement, and **right ventricle** (UBERON:0002080) in the heterozygous form. Secondary involvement: **cardiac atria** (atrial fibrillation), pulmonary/systemic circulation via heart failure. Body system: **cardiovascular system** (UBERON:0004535). In the biallelic syndromic form, additional systems affected: **musculoskeletal system** (skeleton, thoracic cage), **craniofacial structures** (palate), and skeletal/integumentary features (short stature, joint contractures).

**Tissue and cell level.** **Cardiac (striated) muscle tissue** (UBERON:0001133); target cell = **cardiomyocyte** (CL:0000746). Fibrosis reflects **cardiac fibroblast** (CL:0002548) activation and extracellular matrix expansion.

**Subcellular level.** The **sarcomeric M-band** (GO:0031430) is the primary locus of dysfunction, with secondary involvement of the **sarcomere/myofibril** (GO:0030017), and **nucleus** (GO:0005634, reflecting ALPK3's reported nuclear localization and differentiation role).

**Localization / lateralization.** Cardiac involvement is **biventricular/bilateral** with a characteristic **apical predilection** in heterozygotes; hypertrophy may be asymmetric (septal) or concentric/apical.

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## 8. Temporal Development

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**Onset.** - **Biallelic (recessive): congenital/prenatal to neonatal/early-childhood** onset; acute and severe. - **Heterozygous (dominant): adult-onset**, insidious/chronic; onset delayed by ~10 years relative to sarcomeric-gene-positive HCM (PMID: 41645375).

**Progression.** - Biallelic disease is **rapidly progressive**; three founding-cohort patients died of heart failure within the first week of life (PMID: 26846950). A characteristic **neonatal DCM → ventricular hypertrophy transition** occurs in ~44% of live-born biallelic patients (PMID: 32480058). - Heterozygous disease is **slowly progressive/chronic-lifelong**, with left ventricular wall thickness positively correlating with age (notably in female patients).

**Patterns.** Disease is **progressive, not relapsing-remitting**; no spontaneous remission. **Critical windows:** the perinatal period (biallelic lethality) and adult mid-life (penetrance onset in heterozygotes) represent the key vulnerability windows and — for preclinical gene/pharmacotherapy — potential intervention windows.

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## 9. Inheritance and Population

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**Inheritance — dual.** ALPK3 shows **both recessive (biallelic) and dominant (heterozygous truncating)** disease mechanisms, now formally recognized by ClinGen (Finding F010): *"Existing genes were curated for new inheritance patterns where evidence existed"* (PMID: 39971408). ALPK3 carries **established (definitive/strong/moderate) gene-disease validity for HCM**.

**Penetrance & expressivity.** Heterozygous penetrance is **incomplete and age-dependent** — ~20% (**2 of 10 heterozygous family members**) in the founding family (PMID: 26846950); later cohorts suggest even lower penetrance. Expressivity is **variable**, with identical variants producing severe obstructive HCM in one relative and no phenotype in another (PMID: 41645375).

**Epidemiology.** No ALPK3-specific prevalence figure exists. Contextually, **HCM affects ~1 in 500** (PMID: 39971408; PMID: 39132495), and heterozygous ALPK3tv account for ~1–4% of **adult HCM**, giving a rough order-of-magnitude estimate. In apical HCM specifically, ALPK3 is over-represented: **28.6% of genotype-positive apical HCM** in a Swedish cohort (2nd most common after MYH7) (PMID: 40428316).

**Founder effects / consanguinity.** The recessive form was first identified in **consanguineous families** via homozygosity mapping (PMID: 26846950); consanguinity elevates biallelic risk. No specific founder variants are broadly established.

**Sex ratio. Male predominance** among heterozygous ALPK3tv HCM patients (PMID: 41645375).

**Carrier frequency.** Heterozygous truncating ALPK3 variants are present at low frequency in gnomAD; precise carrier frequencies are population-dependent.

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## 10. Diagnostics

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**Imaging (cornerstone).** - **Echocardiography** — detects ventricular hypertrophy, apical morphology, systolic/diastolic dysfunction, outflow obstruction. - **Cardiac MRI (CMR) with late gadolinium enhancement (LGE)** — critical for detecting **apical aneurysm, RV involvement, and fibrosis burden**, all characteristic of ALPK3 HCM, and central to SCD risk stratification (Finding F006). *"Late gadolinium enhancement (LGE) was present in 80% of patients and LGE% independently predi[cted events]"* (PMID: 41759724). Combined markers stratify risk: *"Patients with LVWT  $\geq 30$  mm and LGE  $\geq 15\%$  had a greater risk of SCD (subdistribution hazard ratio, 5.60; 95% confidence interval, 1.90-16.5,  $P = .002$ )"* (PMID: 40317285).

**Electrophysiology.** ECG shows left ventricular hypertrophy patterns (more prevalent in ALPK3tv than sarcomeric HCM); Holter monitoring detects AF and NSVT.

**Genetic testing (definitive for etiology).** - **Whole-exome sequencing (WES)** and **whole-genome sequencing (WGS)** are the primary discovery tools and identified nearly all reported variants. - **HCM/cardiomyopathy gene panels** should **include ALPK3**; historically ALPK3 was

omitted or mis-annotated as recessive-only, causing missed diagnoses — periodic **re-analysis/re-annotation** is essential (PMID: 41645375). - **Single-gene / targeted testing** and **cascade family testing** for known variants; **trio segregation (Sanger)** to establish compound-heterozygous configuration in pediatric cases (PMID: 40447126). - **Chromosomal microarray** may be needed to detect contiguous 15q25 deletions (P 37671554).

**Clinical criteria & differential diagnosis.** Diagnosis follows standard HCM criteria (unexplained LV wall thickness  $\geq 15$  mm, or  $\geq 13$  mm with family history). Differential diagnosis includes sarcomeric HCM (MYH7, MYBPC3), infiltrative/storage cardiomyopathies (Fabry, amyloid), athlete's heart, and syndromic causes (RASopathies) — especially relevant given the extracardiac features of the biallelic form.

**Screening.** Cascade genetic and clinical (imaging) screening of at-risk relatives; prenatal/reproductive counseling for consanguineous families at risk of biallelic disease.

## 11. Outcome / Prognosis

**Biallelic (recessive) form: Poor prognosis** — severe, often lethal in utero, at birth, or in early childhood; heart transplantation, refractory heart failure, and cardiac arrest are reported outcomes; several founding-cohort neonates died within the first week of life (PMID: 26846950; PMID: 38356193).

**Heterozygous (dominant) form: Variable, generally more favorable but with meaningful arrhythmic risk.** In the apical HCM cohort where ALPK3 is prominent, **36.2% experienced HCM-related events**, apical aneurysm occurred in **17.2%**, AF in **41.4%**, and NSVT in **29.3%** (PMID: 40428316).

**Prognostic factors.** LV wall thickness, LGE burden/fibrosis, apical aneurysm, and NSVT are the principal risk markers for SCD. Guideline-based SCD risk algorithms have modest discriminatory power (AUC ~0.58–0.63; PMID: 39713197), and there is **low inter-guideline agreement** for primary-prevention ICD recommendations (Fleiss' kappa 0.340; PMID: 39557320) — reinforcing the value of **CMR-LGE** as an additional stratifier for the fibrosis-prone ALPK3 phenotype.

**Complications.** Heart failure, atrial and ventricular arrhythmias, thromboembolism (AF), apical aneurysm with thrombus, and sudden cardiac death.

## 12. Treatment

There is **no ALPK3-specific approved therapy**; management follows **standard HCM guidelines**, with genotype-directed therapies in preclinical development.

**Pharmacotherapy (guideline HCM care; NCIT terms suggested).** - **Beta-adrenergic blockers** (NCIT:C2496) — first-line for symptoms/obstruction. - **Non-dihydropyridine calcium-channel blockers** (e.g., verapamil). - **Disopyramide** — for obstruction. - Standard heart-failure therapy for the DCM/systolic-dysfunction phenotype. - **Anticoagulation** for AF.

**Genotype-directed / advanced therapeutics (preclinical proof-of-concept; Finding F003).** - **Mavacamten** (cardiac myosin inhibitor; NCIT:C171741) — *"Contractile and calcium handling defects were partly corrected by treatment with mavacamten, a novel myosin inhibitor"* ([PMID: 40128237](#)). Targets the hypercontractility/SRX defect central to ALPK3 pathophysiology. - **AAV-based gene replacement therapy** — *"durable phenotypic rescue in global knockout mice using an adeno-associated virus"* delivering ALPK3 ([PMID: 41221624](#)). Directly addresses the LoF mechanism. - **miR-384-5p modulation** — a candidate therapeutic axis based on amelioration of pressure-overload hypertrophy via the ALPK3 pathway ([PMID: 35510648](#)).

**Surgical / interventional.** - **Septal reduction therapy** — surgical **septal myectomy** (NCIT:C51899) or **alcohol septal ablation** for obstruction. - **Implantable cardioverter-defibrillator (ICD)** (NCIT:C50077) for primary/secondary SCD prevention, guided by CMR-LGE-informed risk stratification. - **Heart transplantation** (NCIT:C15328) for end-stage disease (used in severe pediatric/biallelic cases).

**Supportive care.** Symptom management, exercise counseling, arrhythmia management, and heart-failure supportive therapy.

## 13. Prevention

**Primary prevention.** Because ALPK3 cardiomyopathy is monogenic, primary prevention is **reproductive/genetic**: carrier identification, **genetic counseling** (especially for consanguineous couples at risk of biallelic disease), and reproductive options including **preimplantation genetic testing (PGT)** and prenatal diagnosis.



**Secondary prevention.** Cascade genetic testing and serial clinical/imaging surveillance of at-risk relatives to detect subclinical disease early — important given incomplete, age-dependent penetrance. Periodic re-analysis of previously "negative" HCM genetic tests to capture ALPK3 as knowledge of its dominant mechanism has matured ([PMID: 41645375](#)).

**Tertiary prevention.** Prevention of complications in diagnosed patients: **ICD for SCD prevention** (CMR-LGE-informed), anticoagulation for AF-related thromboembolism, heart-failure management, and activity modification.

**Counseling.** Genetic counseling should address **dual inheritance** — recessive risk in consanguineous unions and dominant, incompletely penetrant risk for offspring of heterozygotes.

No immunization or public-health/environmental prevention is applicable (non-infectious, non-environmental disease).

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## 14. Other Species / Natural Disease

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**Taxonomy & orthologs.** ALPK3 is highly conserved. **Mouse *Alpk3*** (NCBI Gene 116904; *Mus musculus*, NCBI Taxon 10090) is the principal experimental ortholog. Human *ALPK3* is NCBI Gene 57538 (*Homo sapiens*, NCBI Taxon 9606). The gene was originally described as *Midori*, a differentiation-associated cardiac gene.

**Natural disease in other species.** No well-characterized naturally occurring ALPK3 cardiomyopathy in companion animals or wildlife is documented in the reviewed literature (no OMIA entry established in this investigation). Disease knowledge derives from **engineered** rather than spontaneous animal models.

**Comparative biology & conservation.** The **M-band scaffold function and requirement for cardiac function are evolutionarily conserved** — mouse knockouts recapitulate neonatal and adult cardiac dysfunction ([PMID: 40135575](#)), supporting conserved disease mechanisms across mammals.

**Transmission.** Not applicable — genetic, non-zoonotic, non-transmissible.

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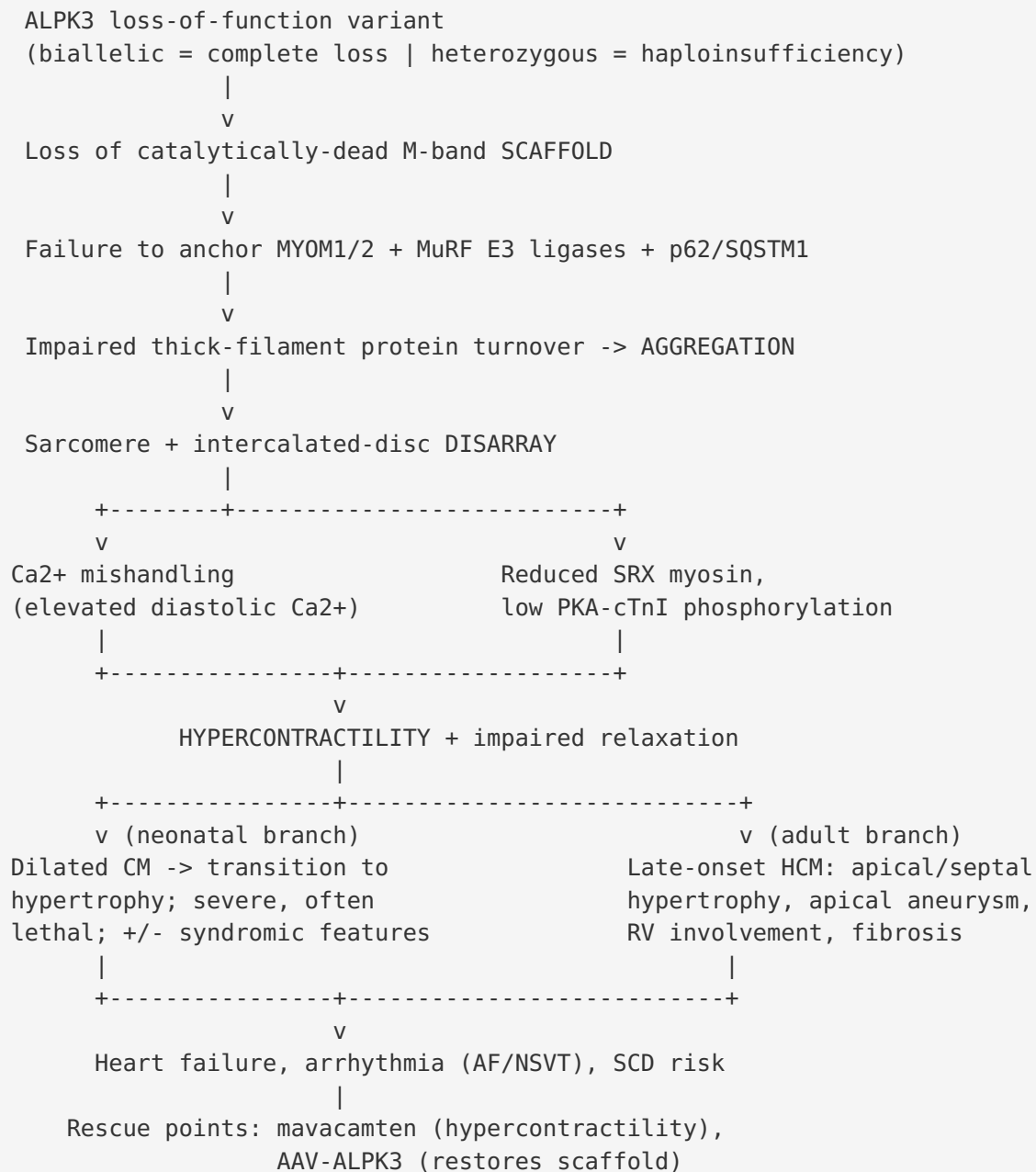
## 15. Model Organisms

ALPK3 cardiomyopathy is supported by a **robust set of complementary models** spanning mouse and human cellular systems.

| Model                                      | Type                       | Key features                                                                                                                 | Reference                                                       |
|--------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <i>Alpk3</i> global knockout mouse         | Mammalian, constitutive KO | Neonatal and adult cardiac dysfunction; scaffold model for thick-filament turnover; <b>AAV-ALPK3 rescue</b>                  | <a href="#">PMID: 40135575</a> ; <a href="#">PMID: 41221624</a> |
| Inducible cardiac-specific <i>Alpk3</i> KO | Mammalian, conditional     | Dissects neonatal vs. adult requirement for ALPK3                                                                            | <a href="#">PMID: 40135575</a>                                  |
| K201X (truncation) knock-in mouse          | Mammalian, knock-in        | Reduced SRX myosin, elevated diastolic Ca <sup>2+</sup> , reduced PKA-cTnI phosphorylation; <b>partial mavacamten rescue</b> | <a href="#">PMID: 40128237</a>                                  |
| Patient-derived iPSC-cardiomyocytes        | In vitro, human            | Disordered sarcomeres/intercalated discs; abnormal calcium handling                                                          | <a href="#">PMID: 27106955</a>                                  |
| ALPK3-mutant human ESC-derived CMs         | In vitro, human            | Establishes ALPK3 deficiency underlies familial cardiomyopathy                                                               | <a href="#">PMID: 27106955</a>                                  |

**Phenotype recapitulation.** Models faithfully reproduce the **core cellular phenotype** (sarcomere disarray, thick-filament aggregation, calcium mishandling, hypercontractility) and the **age-staged cardiac dysfunction**. **Limitations:** models less completely capture the human **syndromic extracardiac features** (facial/skeletal/palate anomalies) and the **age-dependent incomplete penetrance** of the human heterozygous form. **Applications:** mechanism dissection (M-band proteostasis), developmental staging of ALPK3 requirement, and **preclinical testing of mavacamten and AAV gene therapy**.

## Mechanistic Model / Interpretation



The unifying interpretation is that **ALPK3 is a structural (not enzymatic) linchpin of M-band proteostasis**, and disease severity scales with **residual ALPK3 dosage**. This single mechanism parsimoniously explains the entire clinical spectrum — from perinatal lethal recessive disease

to penetrance-limited adult dominant HCM — and rationalizes two mechanistically distinct therapeutic strategies (myosin inhibition to relieve the downstream hypercontractile consequence; gene replacement to restore the upstream scaffold).

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## Evidence Base

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| PMID                                                | Title (abbrev.)                                          | Role in this report                                                                                                       |
|-----------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <a href="#">26846950</a>                            | Biallelic truncating mutations cause severe pediatric CM | <b>Founding study:</b> recessive causation, perinatal lethality, nuclear kinase role, ~20% heterozygote penetrance (F007) |
| <a href="#">41221624</a>                            | ALPK3 Cardiomyopathy: Integrative Review                 | Zygoty–severity relationship, M-band scaffold mechanism, AAV/mavacamten proof-of-concept (F001, F002, F003, F004)         |
| <a href="#">32480058</a>                            | Expanding clinical/genetic spectrum                      | Neonatal DCM→hypertrophy transition (44.4%), adult heterozygous phenotypes (F001)                                         |
| <a href="#">27106955</a>                            | ALPK3-deficient iPSC/ESC cardiomyocytes                  | Ultrastructural disarray, abnormal calcium handling (F002)                                                                |
| <a href="#">40128237</a>                            | ALPK3tv rescued by mavacamten                            | SRX/Ca <sup>2+</sup> defects, partial mavacamten rescue (F003)                                                            |
| <a href="#">40135575</a>                            | ALPK3 essential for neonatal/adult cardiac function      | Mouse KO models, scaffold-for-turnover model (Sections 6, 15)                                                             |
| <a href="#">38356193</a>                            | French multicentric cohort                               | 31 adults heterozygous, HCM main phenotype (26/31), 15% apical/concentric (F004)                                          |
| <a href="#">39606411</a> / <a href="#">40469041</a> | Heterozygous ALPK3tv late-onset HCM                      | Apical involvement and apical aneurysm signature (F004, F008)                                                             |
| <a href="#">30046096</a>                            | Tunisian case, facio-thoraco-skeletal                    | Syndromic biallelic features, specific frameshift variant (F005)                                                          |
| <a href="#">40447126</a>                            | Compound-het pediatric HCM                               | Novel truncating variants, extracardiac features (F005)                                                                   |
| <a href="#">40428316</a>                            | Swedish apical HCM cohort                                | ALPK3 = 2nd most common ApHCM genotype (28.6%), aneurysm 17.2% (F008)                                                     |
| <a href="#">41645375</a>                            | Chinese pedigree re-analysis                             | Dominant inheritance, delayed onset, male predominance, re-annotation importance (Sections 2, 8, 10)                      |
| <a href="#">35510648</a>                            | miR-384-5p protects via ALPK3                            | Post-transcriptional regulation of ALPK3 pathway (F009)                                                                   |

| PMID                                                | Title (abbrev.)                   | Role in this report                                                     |
|-----------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------|
| <a href="#">40317285</a>                            | SCD prediction after myectomy     | LVWT $\geq$ 30 mm + LGE $\geq$ 15% $\rightarrow$ HR 5.60 for SCD (F006) |
| <a href="#">41759724</a>                            | Risk stratification / LGE         | LGE independent SCD predictor (F006)                                    |
| <a href="#">39713197</a>                            | Guideline validation              | Modest guideline discrimination (AUC 0.58–0.63) (Section 11)            |
| <a href="#">39557320</a>                            | Guideline agreement               | Low inter-guideline ICD agreement (kappa 0.340) (Section 11)            |
| <a href="#">39971408</a> / <a href="#">39132495</a> | ClinGen HCM gene reappraisal      | Dual-inheritance curation, HCM ~1 in 500 (F010)                         |
| <a href="#">36660067</a>                            | Overlapping adult-onset phenotype | Restrictive/overlap phenotype, missense variants (Sections 3, 4)        |
| <a href="#">37671554</a>                            | Contiguous 15q25 deletion         | Structural-variant biallelic mechanism (Section 4)                      |
| <a href="#">33076350</a>                            | HCM + skeletal muscle features    | Skeletal muscle involvement (Section 3)                                 |

## Limitations and Knowledge Gaps

- No ALPK3-specific epidemiology.** Prevalence/incidence are inferred from HCM-wide figures (~1 in 500) and the ~1–4% ALPK3 contribution; no direct registry estimate exists.
- Penetrance uncertainty.** The ~20% figure derives from a single founding family; population-scale penetrance of heterozygous ALPK3tv is likely lower and imprecisely quantified. Modifier genes and gene–environment interactions remain unidentified.
- Therapies are preclinical.** Mavacamten rescue and AAV gene therapy are shown only in mouse/iPSC models; no human trials specific to ALPK3 exist. Standard HCM management is extrapolated, not ALPK3-validated.
- Mechanistic gaps.** The relative contributions of ALPK3's nuclear (differentiation) role versus its cytoplasmic M-band scaffold role are not fully resolved, and how catalytic-dead  $\alpha$ -kinase architecture confers scaffolding specificity is incompletely defined.

5. **Extracardiac pathogenesis.** The mechanism linking ALPK3 loss to craniofacial/skeletal malformations in the biallelic form is unexplained.
6. **Variant interpretation.** Historical mis-annotation of ALPK3 as recessive-only led to under-diagnosis; many gene panels and archived tests may still under-call ALPK3.
7. **Model limitations.** Animal/cellular models under-represent the syndromic extracardiac phenotype and age-dependent adult penetrance.

## Proposed Follow-up Experiments / Actions

1. **Population penetrance study.** Leverage large biobanks (e.g., the biobank gene–disease association approach of

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to estimate age-specific penetrance and lifetime HCM risk for heterozygous ALPK3 truncating variants.

2. **Prospective natural-history registry.** Establish an ALPK3-specific longitudinal registry capturing apical aneurysm incidence, arrhythmia burden, LGE progression, and SCD to build an ALPK3-tailored risk model (given weak performance of generic guidelines).
3. **Clinical translation of genotype-directed therapy.** Advance AAV-ALPK3 gene replacement and evaluate mavacamten in ALPK3tv patients via biomarker-driven early-phase trials; define the therapeutic window (neonatal vs. adult).
4. **Modifier discovery.** GWAS/whole-genome and multi-omics analysis of penetrant vs. non-penetrant heterozygous carriers to identify genetic/epigenetic modifiers (including the miR-384-5p axis).
5. **Structural biology.** Determine the ALPK3 M-band interactome structure (with MYOM, MuRF, p62) via cryo-EM/AlphaFold-guided modeling to rationalize truncating-variant effects and design stabilizing therapeutics.
6. **Panel/curation updates.** Ensure all clinical HCM gene panels include ALPK3 with dual-inheritance interpretation, and systematically re-analyze archived "gene-negative" HCM cases.
7. **Mechanism of extracardiac disease.** Use conditional/humanized models to dissect the developmental basis of the syndromic facio-thoraco-skeletal features in biallelic disease.

*Report compiled from 5 investigation iterations, 10 confirmed findings, and 24 reviewed papers. Evidence types: human clinical cohorts and case reports, model-organism (mouse) studies, in vitro human iPSC/ESC-cardiomyocyte studies, and expert-panel gene curation.*

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